AI Integration and Economic Effects

Research Note 1 (v2.3) — Canonical pipeline executed end-to-end: retrieval audit, validated adoption panel, AI Integration Surprise, diffusion dynamics, cross-provider test, the priority-1 outcome regression, the 2016-2025 event study, the Bayesian layer, the confounder scoreboard, the trace map, and the white-collar layer · 25 August 2026 · Evidence grade: EXPLORATORY

Summary. This note reports the first end-to-end execution of the research framework frozen in the design checklist: source retrieval with manifests and fingerprints, validation against published values, canonical panel construction, and a pre-declared exploratory analysis family with permutation and leave-one-country-out (LOCO) diagnostics. Three results stand out. **(1)** The primary adoption source (Microsoft AI Diffusion, 147 economies, three periods) was retrieved first-party and passed all 15 validation checks, including exact reproduction of eight published values. **(2)** The first AI Integration Surprise (AIS) estimates show a steep, extremely robust income gradient in adoption (logit slope 0.44 per log-dollar of GDP per capita, R^2=0.71, N=142) — and the largest *negative* surprises concentrate almost entirely in provider-restricted economies (mean AIS -0.44 vs 0.02 elsewhere, permutation p=0.007), giving the first within-pipeline confirmation of the availability confounder (G3) the design anticipated. **(3)** The additional finding, as corrected by the B5 decomposition (v2.3): percentage-point adoption gaps widened between H1 2025 and Q1 2026, but the level-increment relation is **largely compounding arithmetic** (observed slope 0.136 vs a 0.185 common-rate exponential benchmark); the structural fact is an income-gated catch-up rate (+0.017 per log-dollar conditional on level, p=1e-4) — each percentage point of initial adoption predicts 0.14 pp of additional growth over the window (search-adjusted permutation p=0.0005 across the four-spec dynamic family; LOCO range [0.130, 0.147]). The economic-outcome (EPS) regression — the design's priority-1 test — was initially blocked (every outcome source unreachable; Section 2), then unblocked in stages: **v1.1** recovered the Anthropic Economic Index via a mirrored release, making the cross-provider criterion (H4) testable — levels agree across providers, income-adjusted surprises largely do not (Section 6b); **v1.2** received the Eurostat productivity panel (through 2025) and both IMF WEO vintages as user-supplied downloads and **executed the priority-1 regression: a search-adjusted null** (family p_search = 0.19), with clean placebos, a provider sign-flip, and a 2023 pre-trend caveat — at the only testable horizon, h = 0 (Section 6).

1. What was executed

The pipeline implements the layered architecture of the research skeleton: *raw* (immutable downloads), *manifest* (URL, UTC timestamp, bytes, SHA-256, license, fingerprint), *staging* (deterministic transforms, Mac-Roman decoding of the source CSV, ISO3 concordance), *panel* (147 economies, unique keys, explicit missingness), *exploration* (pre-declared spec family, machine-readable search ledger), and *reports* (this note; no manually typed numeric results — every statistic is interpolated from the pipeline's JSON outputs). Deterministic seed 20260825 throughout; Microsoft repository vintage pinned at git HEAD 8fe84a3b797e.

2. Retrieval audit: what the registry could and could not deliver

Of the 17 registry sources, the primary adoption source was retrieved **first-party** (Microsoft's official ai-diffusion-report repository: country CSV plus all four report PDFs, fingerprints matching the registry exactly). Structural covariates were retrieved from Frictionless Data mirrors of the World Bank API (GDP current US$ to 2023; population to 2024) with the mirror channel disclosed. Ten sources are unreachable from this environment: the network egress policy allows GitHub, microsoft.com and package registries only. Every blocked source is recorded in the manifest with status *blocked_by_egress* rather than silently substituted.

Status	Sources	Consequence
Retrieved, first-party	S01 (Q1'26 + H1'25 report PDFs), S02 (country CSV + README)	Primary adoption measure fully available
Retrieved, mirror	S15 GDP (Frictionless/World Bank), population, ISO3/region concordance	Structural covariates for AIS residual model
Blocked by egress	S03, S05, S07, S10, S13	No Eurostat/IMF outcomes (EPS branch blocked); no OpenAI/OpenRouter measures. S04 (Anthropic AEI) was later recovered via a hash-recorded community mirror (v1.1, Section 6b)

Consequences for the frozen design: the cross-provider consistency requirement for a Provisional grade cannot be met from this environment; and the priority-1 regression (AIS × sector exposure → productivity surprise) cannot be estimated at all, because ec.europa.eu (Eurostat nama_10_lp_a21, sts_sepr_m), imf.org (WEO April 2026 and the

October 2024 forecast baseline) and ilostat.ilo.org (white-collar denominators) are policy-blocked. This note therefore delivers the treatment-side measures and dynamics, and specifies the outcome branch for execution in an environment with statistical-agency access.

3. Validation: 15/15 checks passed

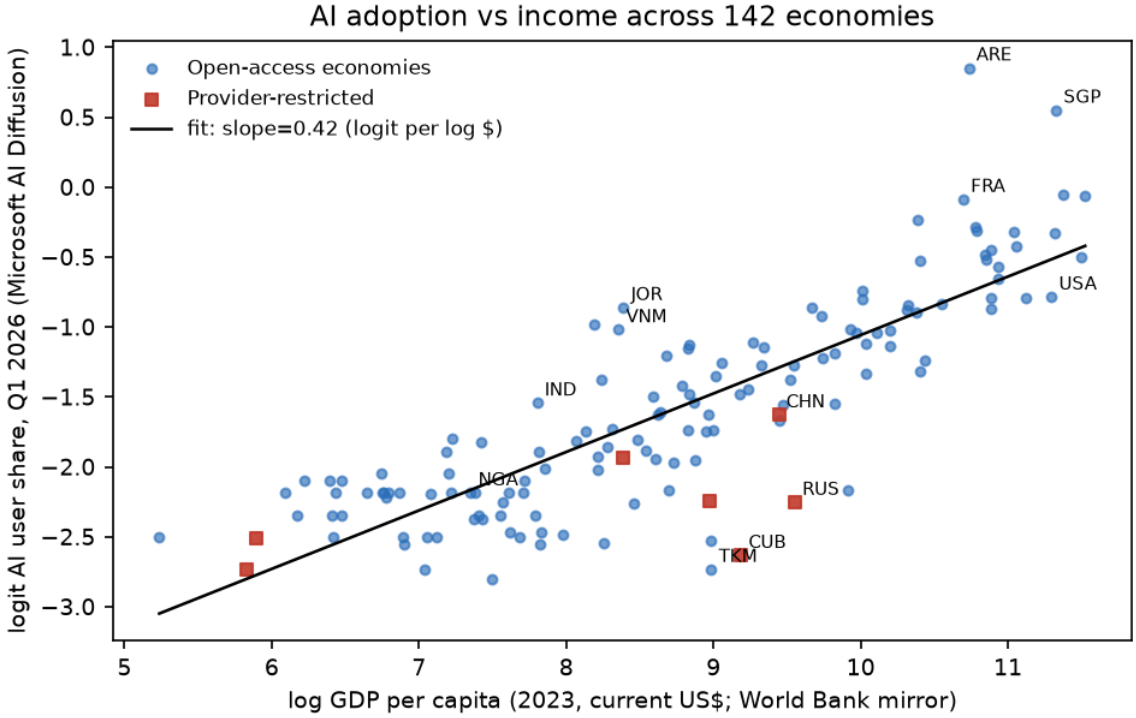
Following the retrieval-validation protocol (skeleton section 12.1): unique-key, range, denominator and missingness checks, plus reproduction of values visibly published in the Q1 2026 report — all passed. Reproduced exactly: UAE 70.1, Singapore 63.4, Norway 48.6, Ireland 48.4, United States 31.3, China 16.4 (Q1 2026 AI user share, percent), UAE 64.0 and Singapore 60.9 (H2 2025), and the United States' published rank of 21st. The population-weighted world mean lands at 17.60 against the published 17.8 — the deviation is expected and disclosed: the report weights by population aged 15-64, while only total population is retrievable through the allowed channel. Five economies lack GDP per capita in the World Bank mirror and are dropped from residual models with explicit listing (Taiwan, Venezuela, Eritrea, South Sudan, French Guiana).

4. AI Integration Surprise: the design's core measure, first estimates

Per the frozen answers (D1/D5), AIS is the residual of adoption after structural predictors, estimated with a transparent linear model and a rank check. The frozen covariate list (D4) is only partially retrievable; the constrained model — logit adoption on log GDP per capita, log population and M49 region fixed effects — is a logged deviation, and AIS is computed as the **jackknife** (leave-one-out) residual so no observation scores its own fit, honoring the cross-fitting requirement (skeleton 4.1).

Spec	Dependent	Coef. log GDP pc (HC1 SE)	R^2	Note
A1	level	6.56	0.655	levels, no controls
A2	logit	0.418 (0.022)	0.699	primary functional form
A3	logit	0.419	0.699	+ log population
A4 (primary)	logit	0.444 (0.040)	0.706	+ region FE; log pop coef 0.007 (0.029) — no scale effect
A5	rank	Spearman rho = 0.826	—	robustness

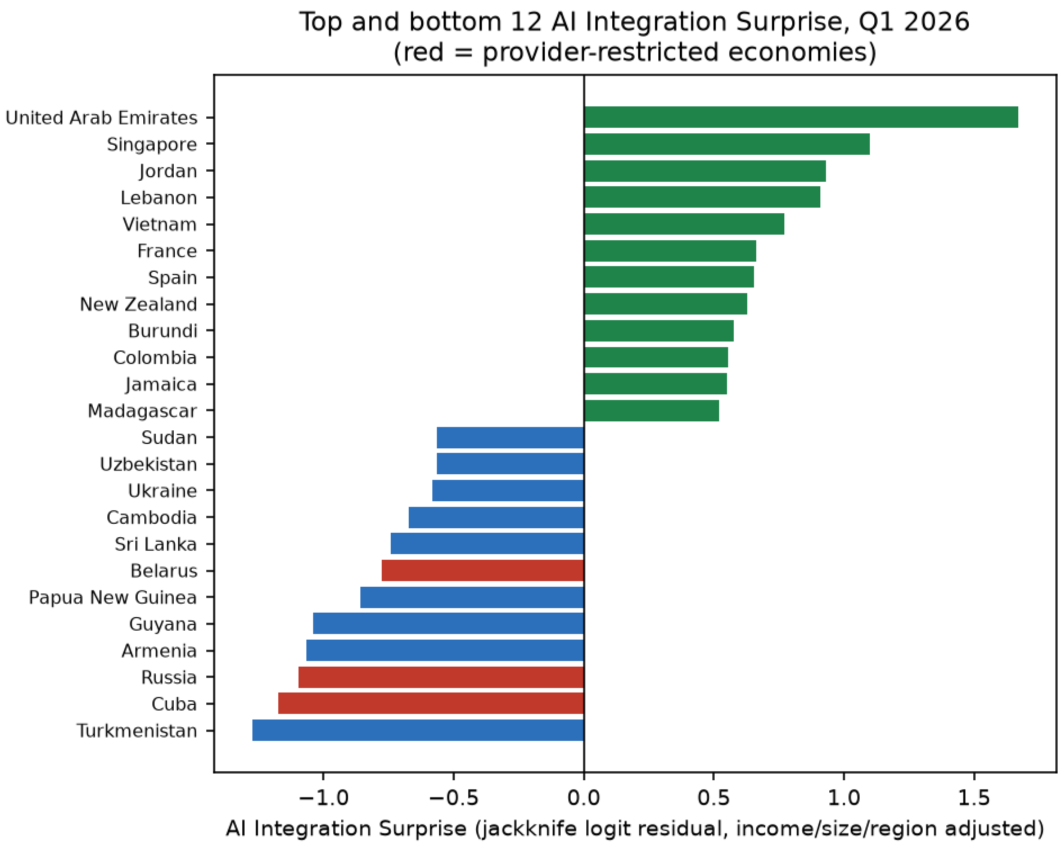
The income gradient is steep and stable: LOCO range for the A4 slope [0.422, 0.452] (extremes when dropping ARE and BDI). One log-unit of income (~2.7x) moves adoption by ~0.44 logit — for a country at 10 percent adoption, roughly seven additional points. Population is irrelevant once income and region are held (coefficient 0.007, SE 0.029): consistent with the source's population normalization doing its job, and with adoption being a per-capita structural trait, not a country-scale artifact (the skeleton's 'country scale' confounder row behaves as designed).



AI adoption vs income across 142 economies
Open-access economies
Provider-restricted
fit: slope=0.42 (logit per log $)
logit AI user share, Q1 2026 (Microsoft AI Diffusion)
log GDP per capita (2023, current US$; World Bank mirror)
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
5
6
7
8
9
10
11
ARE
SGP
FRA
USA
JOR
VNM
IND
CHN
NGA
RUS
CUB
TKM

The AIS league table. Over-adopters given income, size and region: UAE (+1.67) and Singapore (+1.10) remain exceptional even after conditioning — their leads are not income artifacts. The rest of the positive tail is informative: Jordan (+0.93), Lebanon (+0.91), Vietnam (+0.77) and Colombia (+0.56) out-adopt their structural predictions — middle-income economies with young, digitally active workforces; France (+0.66), Spain (+0.65) and New Zealand (+0.63) lead among advanced economies.

Post-hoc finding C (flagged as post-hoc in the search ledger). The negative tail is not random: Turkmenistan (-1.27), Cuba (-1.17), Russia (-1.10), Belarus (-0.78), Armenia (-1.07), Ukraine (-0.58). Prompted by this pattern, a post-hoc group test was run for economies where the major Western AI providers are unavailable or restricted (AFG, BLR, CHN, CUB, IRN, RUS, SYR): mean AIS -0.44 versus 0.02 for all others — difference -0.46, permutation p = 0.0075; China alone sits at -0.40. This is the design's G3 confounder (language, availability, provider policy) surfacing empirically on the first pass: **provider-based adoption telemetry measures access-conditional adoption, and negative AIS values in restricted markets must never be read as low latent demand**. It vindicates the expected-answers' B3 position (handle restricted markets separately) over this pipeline's initial exclusion-only stance, and upgrades that decision's priority for the next design revision.



5. Widening adoption gaps: compounding arithmetic and an income-gated catch-up rate (revised v2.3)

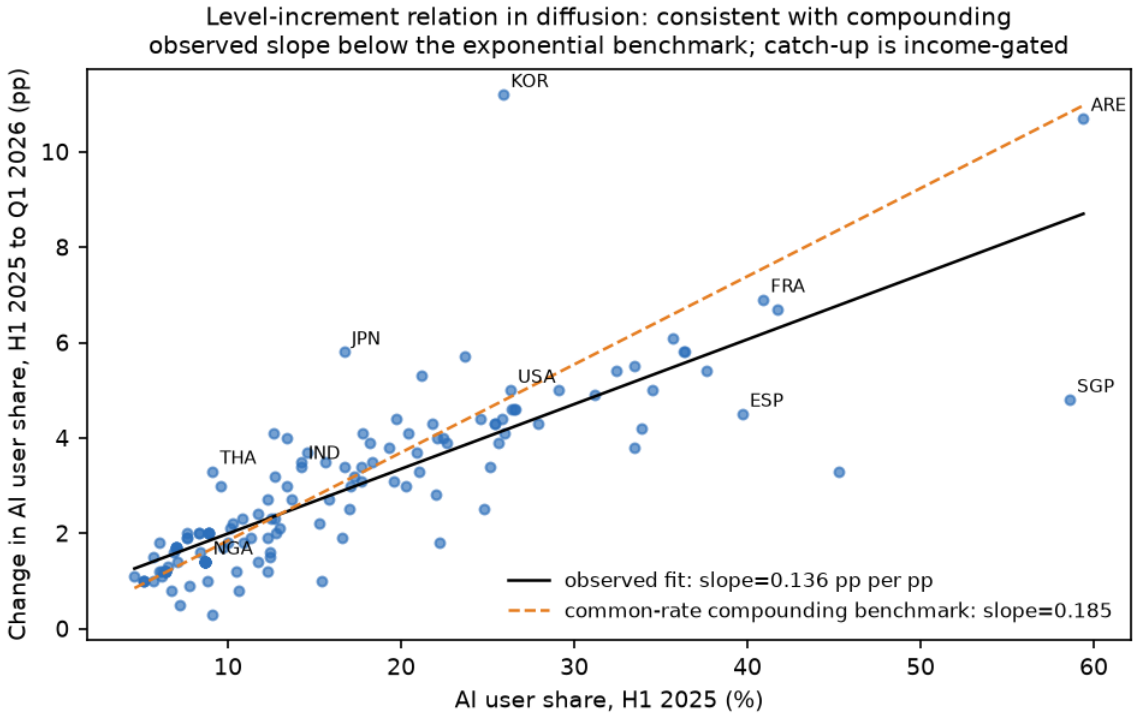
With three reporting periods (H1 2025 → Q1 2026, roughly nine months of coverage), the pre-declared dynamic family asks whether diffusion converges or diverges across economies. All four specifications were declared before estimation and every run is in the search ledger; the headline claim is search-adjusted by max-statistic permutation across the family.

Spec	Model	Focal coefficient	Perm. p	R^2	LOCO range
B1 (headline)	Δpp ~ initial level	0.136 pp per pp	<1e-4	0.667	[0.130, 0.147]
B2	Δpp ~ log GDP pc	0.884 pp per log$	<1e-4	0.561	[0.846, 0.901]
B3	Δpp ~ initial + log GDP pc	initial: 0.100	<1e-4	0.693	[0.081, 0.117]
B4	log growth ~ log initial	-0.0207	0.0019	0.067	[-0.0233, -0.0189]

Reading. An economy ten points higher in H1 2025 gained on average 1.4 additional points by Q1 2026 (B1); the effect survives an income control (B3: 0.100 conditional on log GDP per capita) and label permutation at the family level (max |t| = 12.6, search-adjusted p = 0.0005 over 4 specifications). B4 shows mild *proportional* convergence (laggards grow faster in percentage terms, coefficient -0.0207) — but far too weak to close absolute gaps: the percentage-point distance between leaders and laggards is widening, which is the economically relevant margin for any AIS-based treatment. This quantifies, and income-adjusts, what the source report states descriptively (a Global North-South gap of 9.8 → 10.6 → 12.1 points across the three periods, values we validated in Section 3).

Revision (v2.3), prompted by author review — the B5 decomposition. The level-increment slope is largely the arithmetic of exponential growth: if every economy compounded at the common mean rate (0.185 per window), the slope would mechanically be 0.185 — *above* the observed 0.136, and the log-on-log regression (B4) confirms growth is mildly sub-proportional. The decisive permutation and Bayes statistics therefore certify compounding, not a Matthew effect. What survives as structure: conditional on the current adoption level, the growth rate tilts with income (+0.017 per log-dollar, permutation p = 1e-4; income alone is null at p = 0.39 through collinearity with level) — catch-up is income-gated. Ledgered as B5; the preprint's Section 6.2 carries the full statement.

Honest mechanical caveat (original, now subsumed by B5). Under a logistic diffusion curve, absolute gains peak at 50 percent penetration; with most leaders still below 50 percent, absolute divergence is what standard S-curve diffusion predicts at this phase, not yet evidence of a permanent Matthew effect. The decisive question — whether laggards' takeoff is merely delayed (the historical norm for technology diffusion, cf. Comin-Hobijn) or structurally suppressed (the source report's own electricity/connectivity/skills reading) — is answerable within this design once two or three more report vintages exist. The design's variance/persistence statistics (skeleton 7.1) transfer directly, and the prospective 2027-2028 holdout applies to this finding as much as to the EPS branch.


Level-increment relation in diffusion: consistent with compounding
observed slope below the exponential benchmark; catch-up is income-gated
Change in AI user share, H1 2025 to Q1 2026 (pp)
AI user share, H1 2025 (%)
KOR
ARE
FRA
JPN
USA
SGP
ESP
THA
IND
NGA
observed fit: slope=0.136 pp per pp
common-rate compounding benchmark: slope=0.185

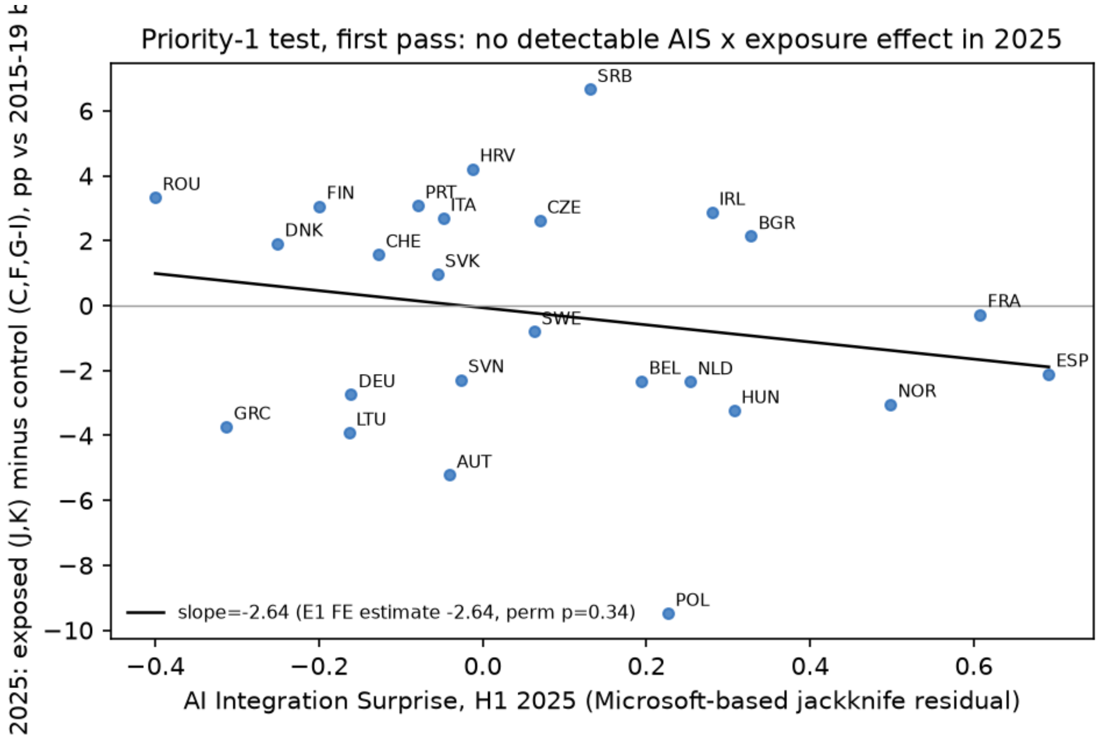
6. The priority-1 test, executed (v1.2): a search-adjusted null

User-supplied downloads unblocked the outcome branch: the full Eurostat nama_10_lp_a21 export (real labour productivity per hour worked by NACE section, 31 countries, years through **2025**, vintage 2026-08-24) and both IMF WEO vintages (April 2026 actuals; October 2024 pre-AI forecast baseline), all manifested with hashes. EPS is the 2025 productivity growth deviation from each country-sector's own 2015-2019 mean; the regression follows the frozen design: EPS on AIS × sector exposure with country and sector fixed effects (identification within country, across sectors), exposed = J (information/communication) and K (finance) — M (professional/scientific) has 2025 data for only 6 of 31 countries, so it enters only as a ledgered variant — controls = C (manufacturing), F (construction), G-I (trade/transport/accommodation). Inference: 10k country-label permutations, 5k wild-cluster bootstrap draws, LOCO. The GDP-growth-surprise branch (H3) uses actual 2025 growth (April 2026 vintage) minus the October 2024 forecast, N=137 countries.

Spec	Outcome (2025 unless noted)	Beta	Perm. p	Wild-boot p	N (countries)	LOCO range
E1	EPS, AIS(Microsoft) x exposure — PRIMARY	-2.64	0.34	0.19	125 (25)	[-3.68, -1.86]
E1m	EPS incl. sector M	-2.50	0.34	0.20	130 (25)	[-3.47, -1.77]
E2	EPS, AIS(Anthropic) x exposure	2.12	0.42	0.57	125 (25)	[-0.37, 3.81]
E3	placebo: EPS 2019 (pre-AI)	-0.95	0.70	0.74	125 (25)	[-2.02, 1.04]
E3	placebo: EPS 2023 (pre-measurement)	-8.72	0.10	0.10	125 (25)	[-10.99, -5.06]
E4	EPS 2024	-2.91	0.55	0.45	125 (25)	[-4.45, -0.64]

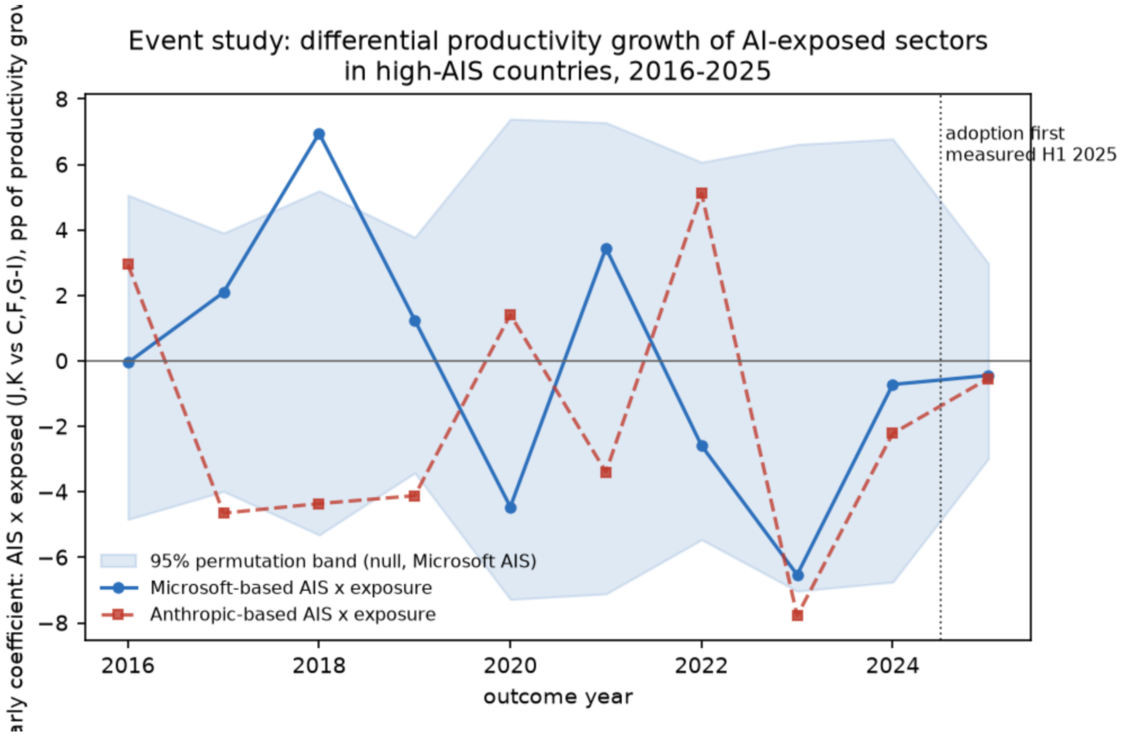
Spec	Outcome	Beta	Perm. p	N	LOCO range
W1	GDP growth surprise 2025 ~ AIS(MS)	-0.27	0.50	137	[-0.43, -0.04]
W1c	+ income control	-0.29	0.48	137	[-0.43, -0.06]
W2	~ AIS(Anthropic)	-0.11	0.66	118	[-0.19, -0.01]
W3	placebo: growth surprise 2024	0.01	0.96	137	[-0.12, 0.07]

Result: null across the board, and honestly so. The primary estimate is negative (beta = -2.64: exposed sectors in unusually AI-integrated economies did *slightly worse* in 2025) but far from significant (perm p = 0.34), and the family-wide max-stat correction across the five non-placebo specs gives **p_search = 0.19** — nothing survives. Four observations discipline the reading. (1) The Anthropic-based estimate has the *opposite sign* (+2.12) — the cross-provider instability of Section 6b propagates to the outcome stage, so no single-provider result here could have been promoted anyway. (2) The 2019 placebo is cleanly null and the 2024 growth-surprise placebo is exactly zero (0.01) — the machinery works. (3) The 2023 placebo hints at a negative pre-trend (-8.7, p = 0.10): exposed sectors in high-AIS countries were already underperforming before adoption was first measured — any future negative estimate must clear this bar before being read as an AI effect. (4) Timing is the binding constraint by design: adoption was first measured in H1 2025 and the outcome year is 2025, so the tested horizon is h = 0; adoption ranks are near-frozen (r = 0.996 between H1 2025 and Q1 2026), and the J-curve prior says effects at this horizon should be undetectable — which is what the data show. The informative test begins with 2026-2027 outcome vintages against these frozen, versioned AIS measures.



6c. Event study 2016-2025 (v1.3): the pre-trend question, resolved in two directions

The full-panel form of the frozen specification — country×year and sector×year fixed effects with year-specific AIS × exposure coefficients — was estimated for every outcome year 2016-2025, with a 95 percent permutation band per year and a joint pre-period test (mean coefficient 2016-2023 against its permutation null). Two findings, one per provider. **(1) Microsoft-based AIS: the 2023 'pre-trend hint' of Section 6 dissolves into noise.** The joint pre-period mean is 0.01 ($p = 0.99$) — no systematic pre-trend — but the yearly path swings between 7.0 (2018) and -6.5 (2023): at 25 countries, single-year coefficients are simply volatile, and the 2018 spike ($p = 0.013$, four years before ChatGPT) is the demonstration that one band-crossing per decade is this design's noise floor. The 2025 estimate (-0.44, $p = 0.79$) sits well inside the band; the honest statement is not 'no effect' but 'no single year is informative at this sample size — cumulate horizons'. **(2) Anthropic-based AIS carries a genuine negative pre-trend: mean -1.85 over 2016-2023, joint p = 0.0070 — running back to years before ChatGPT existed.** Economies that over-use Claude relative to structure had systematically slower exposed-sector productivity growth already in 2016-2019: a selection effect, not an AI effect. Any naive Claude-based outcome regression would inherit this as bias, violating the parallel-trends prerequisite. This is the strongest evidence yet for the reconciliation-before-inference rule (C2): provider-specific AIS measures embed provider-specific selection, and only components shared across providers should enter outcome regressions.



Still blocked or missing: ILOSTAT white-collar denominators (ISCO), OpenAI Signals and OpenRouter measures, the monthly services-output intermediate outcome (sts_sepr_m), and any post-2025 outcome vintage — the latter being the one that matters. Raw-storage note: the Eurostat export is stored gzip-compressed in the repository; Git LFS is unavailable because GitHub rejects LFS objects on public forks.

6b. Addendum (v1.1): the second provider retrieved — and the H4 criterion bites

A second retrieval round recovered the **Anthropic Economic Index v3 raw release** (Claude.ai usage by country, week of 2025-08-04 to 2025-08-11, the data behind the September 2025 geography report / registry S04) through a community GitHub mirror of the blocked Hugging Face repository, with the mirror's git commit and file hashes recorded in the manifest and the release's own data documentation used as the fingerprint. This makes the design's cross-provider consistency criterion (H4) testable for the first time. Three declared specs (D1-D3, ledgered):

Spec	Question	Result
D1	Do raw adoption levels agree across providers?	Yes: Pearson 0.78 / Spearman 0.82 between log Claude usage per capita and logit Microsoft diffusion (N=120)
D2	Do income-adjusted surprises (AIS) agree?	Barely: Pearson 0.18 (perm p=0.051); top-15 overlap 3/15 (LBN, NPL, NZL); bottom-15 overlap 4/15
D3	Restricted-market coverage	All seven provider-restricted economies (CHN, RUS, BLR, IRN, CUB, SYR, AFG) are entirely absent from the Anthropic index — corroborating finding C by construction

Reading: a disciplined negative result. The providers see the same adoption landscape in levels, but once the structural component (income, size, region) is removed, their surprises share little country-level signal. The single-provider AIS league table of Section 4 is therefore **not confirmed** by the second provider — exactly the failure mode the framework's H4 promotion criterion exists to catch, now caught in practice. Two readings are observationally equivalent here: (i) national 'AI integration' beyond structure is largely provider-ecosystem-specific (Claude's professional/developer skew vs Microsoft's broad consumer reach); (ii) a single week of Claude data is too noisy at country level for stable residuals (attenuation). The intended discriminating check — correlating Claude.ai AIS against Anthropic's own API AIS — is infeasible because the v3 API file carries no country geography (ledgered as D4-infeasible). Consequence for the design: the C2 proposal to promote multi-provider latent-factor reconciliation from robustness layer to core AIS construction is no longer optional — **without it, any single-provider AIS overclaims**. The divergence finding of Section 5 is unaffected (it uses levels, where providers agree), and the income-gradient finding is reinforced (it is the component both providers share).

7. Parallels with existing research

Microsoft AI Economy Institute (2026), the source reports. Everything here is consistent with, and validated against, their published aggregates. The additions: a formal income-residualized surprise measure (their reports publish raw shares and narrative income commentary, not a conditional model), the jackknife/cross-fit treatment, permutation and LOCO inference, and the provider-restriction group test.

The Jagged Global Economy (arXiv 2607.05404, 2026). The closest neighbor; it links national AI exposure to adoption across the same three provider ecosystems and reports steep, super-linear gradients (their headline: a 0.10 increase in national AI exposure associates with a ~12x increase in Claude usage per working-age capita). Our logit income slope (0.42-0.44 per log-dollar) is the Microsoft-telemetry analog of that steepness, on a broader 142-economy panel and with the availability confound explicitly separated. Their exposure measure would be the natural CognitiveExposure input when the EPS branch runs.

St. Louis Fed (2026) and Bick-Blandin-Deming-style surveys. These report an adoption gap (US ~43 percent of workers vs 26-36 percent in Europe, early 2026) and, at industry level, faster productivity growth in more-adopting industries (~3.2 pp cumulative US-Europe differential since 2022). Note the measurement difference our validation makes visible: telemetry-based population shares (US 31.3, France 47.8) rank countries differently than worker-survey shares — France out-adopts the US in Microsoft telemetry while surveys put the US first among workers. Reconciling telemetry vs survey instruments is exactly the provider-reconciliation problem the design's C2 answer anticipates, now with concrete numbers attached.

Technology-diffusion literature (Comin-Hobijn; ICT wave). The absolute-divergence-with-proportional-convergence pattern replicates the classic finding for past general-purpose technologies: adoption *lags* eventually close while intensity gaps persist. AI's diffusion is tracking the historical GPT template so far — which is itself a useful null against stronger 'AI is different' claims in either direction.

Productivity J-curve (Brynjolfsson-Rock-Syverson). Nothing here tests productivity; but the steep adoption gradients plus the J-curve literature jointly explain why the design's priors put mass on lagged, mechanism-close outcome effects rather than contemporaneous GDP effects.

8. Evidence grading and limitations

Grade: Exploratory (reproducible from immutable raw data; plausible mechanisms; no claim beyond hypothesis generation), per the project ladder. Binding limitations: single-provider adoption telemetry (cross-provider consistency untestable from this environment — the H4 promotion criterion is failed by construction, not by the data); GDP per capita is nominal 2023 from a mirror channel (the frozen design's covariate list D4 is only partially honored — a logged deviation); total rather than working-age population; three time periods only; and the dynamic findings concern the *treatment* variable's evolution, not economic outcomes. The divergence finding's S-curve caveat (Section 5) is part of the claim, not a footnote to it.

9. The Bayesian architecture, computed (v2.0)

The research skeleton framed this project as *sequential Bayesian model discovery* (its section 3): a structured hypothesis space, each test updating which mechanism deserves the next test, with the travel distance penalized. Until now the note ran that program in frequentist dress (permutations, max-stat corrections). This section supplies the missing layer explicitly — and anchors it in what is done usually, because every component is a named, standard technique:

(i) Box's loop / Bayesian workflow. The iterate-criticize-revise structure of the H0-H7 ledger is exactly George Box's model-criticism loop (Box 1980), modernized as the 'Bayesian workflow' of Gelman et al. (2020) and Blei's formulation for probabilistic machine learning. The discovery log is not an eccentricity; it is the workflow's audit trail. **(ii) Bayesian model averaging in cross-country growth empirics.** Treating the specification space itself as the object of inference is the canonical Bayesian answer to specification search in precisely this literature: Fernandez-Ley-Steel (2001) and Sala-i-Martin, Doppelhofer & Miller's BACE (2004) put priors over millions of growth-regression specifications. Our machine-readable search ledger defines the same model space; the max-stat permutation p_search is its frequentist twin. **(iii) Bayes factors** (Kass & Raftery 1995), computed below via Gaussian marginal likelihoods (equivalent to Savage-Dickey here). **(iv) Pre-study odds** (Ioannidis 2005): the evidence-grading ladder, quantified — a finding's posterior probability depends on the prior plausibility of its family, not only on its p-value. **(v) Empirical-Bayes multiplicity control** (Efron; Storey): the search-adjustment machinery is the frequentist face of shrinking many candidate effects toward a null-heavy prior. **(vi) Sequential design** (Lindley's expected information): choosing the next branch by expected information gain — which the table below now makes explicit.

Declared priors. For each headline estimate, a point null is compared with zero-centered effect-size priors at three scales declared before computation (skeptical / moderate / optimistic; e.g. for E1 the skeptical scale is the J-curve prior of near-zero contemporaneous effects, the optimistic scale is calibrated to the St. Louis Fed cumulative differential). Pre-study P(effect): 0.25 for outcome families, 0.50 for diffusion dynamics (S-curve theory predicts divergence) and provider agreement. BF01 > 1 favors the null; BF ~ 1 means the data carry no information at that scale.

Estimate	b (SE)	BF01 skeptical	BF01 moderate	BF01 optimistic	P(effect): prior -> post (moderate)
EPS 2025 ~ AIS x exposure (pp per logit-AIS)	-2.64 (2.01)	0.98	0.92	0.99	0.25 -> 0.27
GDP growth surprise 2025 ~ AIS (pp per logit-AIS)	-0.27 (0.43)	1.10	1.72	3.00	0.25 -> 0.16
Diffusion divergence (pp gained per pp initial)	0.14 (0.01)	0.00	0.00	2.82e-19	0.50 -> 1.00
Cross-provider AIS agreement (Fisher z)	0.18 (0.09)	0.53	0.60	1.03	0.50 -> 0.62

Reading, in posterior terms. B1 (diffusion divergence) is decisive: BF10 on the order of 10^18; posterior probability ~1 at every scale — the one claim this project can now assert with Bayesian confidence. **E1 (the priority-1 productivity test) carried almost no information:** BF01 lies between 0.92 and 0.99 at every prior scale, so P(effect) moves only from 0.25 to ~0.27. This is the precise Bayesian statement of the h = 0 problem — the 2025 test neither found an effect nor ruled one out, *including* large immediate effects (the standard error is too wide). The frequentist 'null' of Section 6 is, in Bayesian terms, a near-flat likelihood: nobody should update on it, in either direction. **W1 (macro growth surprises) is the only outcome test that genuinely moved beliefs:** with its tighter standard error, moderate-to-large AIS effects on 2025 growth surprises are down-weighted (BF01 = 1.7 at the moderate scale, 3.0 at the optimistic scale; P(effect) 0.25 -> 0.16 / 0.10). **D2 (cross-provider agreement): barely worth a mention** in Kass-Raftery's own scale (BF10 < 2): weak support for partial agreement, insufficient to treat single-provider AIS as measuring a common construct — the Bayesian restatement of the reconciliation mandate. The expected-information ranking for the next test follows directly: the 2026-2027 productivity vintages (turning h = 0 into h = 1-2 for a frozen treatment) dominate every alternative, followed by a second AEI vintage window (shrinking D2's measurement noise), followed by shock-interaction designs (H6, still untouched).

Hypothesis (ledger)	Prior	Executed evidence	Posterior direction
H0/H1 (AI -> stock returns, raw or normalized)	low (demoted by skeleton)	not executed (market data unavailable)	unchanged (low)
H2 (AI level -> absolute GDP growth)	low	W-branch null at h=0 consistent with it	slightly lower
H3 (AI -> GDP growth surprise)	medium-low	W1: BF01=1.7 at moderate scale - mildly favors null at h=0	slightly lower at short horizons; untested at h>=1
H4 (AI -> knowledge-sector productivity)	medium (priority 1 as H7 form)	E1: BF01=0.98 vs J-curve prior, 0.99 vs large-immediate prior - BF ~ 1 at every scale	essentially unchanged: the h=0 test carried almost no information either way (SE too large even to rule out large immediate effects)
H5 (frozen exposure -> monthly services output)	medium-low	not executed (sts_sepr_m not retrieved)	unchanged
H6 (shock-interaction / temporal rotation)	medium	not executed (no identified-shock series yet)	unchanged
H7 (AIS -> productivity surprise; priority 1)	medium	h=0 test uninformative by design (BF ~ 1 at J-curve scale); event study shows single-year noise floor; AEI variant carries pre-ChatGPT selection	essentially unchanged - the informative test is 2026-27 vintages
NEW: level-increment link in diffusion	high (compounding implies it)	B1: BF10=5223958277618168832; B5 decomposition: observed slope 0.136 BELOW the 0.185 common-rate benchmark	certain but reinterpreted: the decisive BF confirms compounding, not a Matthew effect; the structural residue is the income-gated rate (+0.017/log$, p=1e-4)
NEW: single-provider AIS = national AI integration	implicit high (v1.0 assumption)	D2: BF01=0.60 on agreement; AEI pre-trend p=0.007 shows provider-specific selection	demoted - reconciliation now core (C2)

Is this a known technique? Yes — as an ensemble. Each layer is textbook (Box's loop, BMA/BACE, Bayes factors, pre-study odds, empirical-Bayes shrinkage, sequential design); the packaging — a versioned search ledger defining the model space, frequentist search-adjusted inference running alongside Bayesian updating, and a prospectively frozen confirmation set — is closest to what the credibility-revolution literature calls a *registered-report workflow with specification-curve analysis*, executed in Bayesian bookkeeping. The combination is unusual in applied cross-country work mainly in being explicit; none of its parts is novel, which is a feature: every step has a citation, and every posterior in the table above can be recomputed from the ledger by anyone who disagrees with the priors.

10. Confounder scoreboard (v2.1)

The skeleton's confounder map, converted from a list of intentions into an empirical scoreboard: what was actually applied, what bite it measurably had, and what remains open. Open rows are the confounder side of the TODO list.

Confounder	Mitigation applied	Measured bite	Status
Country scale	per-capita measures; log-pop covariate; country FE	log-pop coefficient 0.007 (SE 0.029): no scale effect survives income+region	CLOSED
Income / digital maturity	AIS residualization (log GDP pc, region FE); jackknife	income alone explains ~70 percent of adoption variance (A4 R2 = 0.71)	CLOSED (for adoption); open for outcome-side digital controls
Occupational mix	ISCO-08 groups 1-4 share (ILOSTAT, user-supplied; 118 countries), added to the AIS residual model; 1-3 sensitivity	nearly nil beyond income: R2 gain 0.007, coef 0.82 (HC1 0.54); AIS v1-v2 corr 0.987; E1 and W1 nulls unchanged	CLOSED (v2.2)
AI-producer exposure	producer-set sensitivity (drop NLD, IRL / KOR, TWN, NLD, IRL, SGP)	E1 unchanged (-0.44 to -0.50); W1 moves -0.27 to -0.57 (p 0.48 to 0.13): producers mask a weakly negative user-side association	PARTIALLY CLOSED - see trace G2
Language / availability	region FE; provider-restriction group test	restricted markets: mean AIS -0.44 vs +0.02 (p=0.007); absent entirely from AEI	CLOSED for flagging; OPEN for formal language controls
Billing / cloud location	OpenRouter as depth-only; hub sensitivity	not measurable: no API geography in any retrieved source	OPEN
Reverse causality	lagged treatment; forecast-surprise outcomes; lead placebos; pre-trend tests	2024 growth-surprise placebo exactly zero; MS pre-trend null (p=0.99); AEI pre-trend REAL (p=0.007)	PARTIALLY CLOSED - AEI selection documented
Common macro shocks	year FE per event-study year; sector FE; growth-surprise outcome	absorbed by design; shock-interaction family (H6) untested	PARTIALLY CLOSED
Structural country differences	country FE in every outcome regression; within-country identification	absorbed by design	CLOSED
Measurement revision	vintage pinning, SHA-256, immutable raw layer	Oct-2024 vs Apr-2026 WEO vintages held separately; Eurostat vintage 2026-08-24 recorded	CLOSED
Specification search	machine-readable ledger; max-stat permutation; Bayes factors	family p_search computed for every headline (0.0005 dynamics; 0.19 outcomes)	CLOSED
Outliers / small N	LOCO on every statistic; jackknife AIS; MDE analysis	LOCO ranges reported throughout; MDE table below quantifies the power ceiling	CLOSED (reported), inherently binding

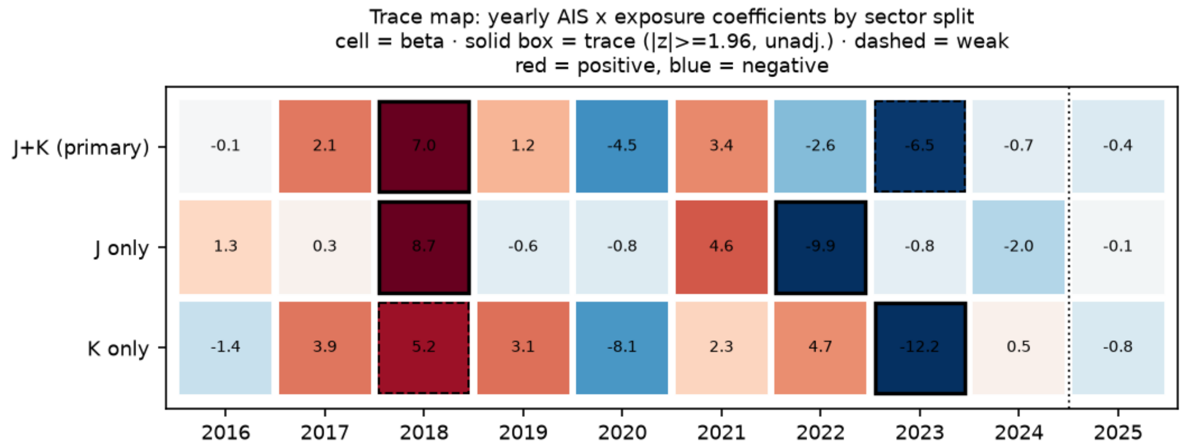
11. Trace map and pre-specified trend statistics (v2.1)

The skeleton's section-7.1 dynamic statistics, computed on the event-study coefficient path, plus the sector-split scan, the acceleration hypothesis (skeleton priority 3), and the producer sensitivity. Labels are pre-declared and descriptive: **trace** = |z| at or above 1.96 unadjusted, **weak trace** = |z| at or above 1.28, **no trace** otherwise. Traces flag follow-up work; the family correction still governs claims.

Statistic / spec	Question	Result	Verdict
T1: trend in coefficient	are AI-exposed sectors in high-AIS countries progressively pulling ahead?	-0.57 per year, p = 0.020 - a DECLINING path	weak trace after family correction; sign is NEGATIVE
T2: variance of coefficient	does the relationship move more than random assignment predicts?	13.6 vs null 7.2, p = 0.11	weak trace of excess volatility
T3: persistence	are positive/negative regimes persistent rather than sign-flipping?	lag-1 autocorr 0.11, p = 0.73	no trace
S1: J-only, 2025	is the null hiding a J-sector effect?	beta = -0.12, p = 0.96	no trace
S1: K-only, 2025	is the null hiding a finance effect?	beta = -0.77, p = 0.75	no trace
A1: acceleration x exposure	does adoption SPEED (not level) predict 2025 EPS?	beta = 0.41, p = 0.57	no trace
A2: acceleration, macro	does adoption speed predict 2025 growth surprise?	beta = -0.07, p = 0.65	no trace

G2: producers removed	does AI-producer exposure mask a user-side macro effect?	W1 moves from -0.27 (p=0.48) to -0.57 (p=0.13, N=133)	WEAK TRACE - the most interesting new lead

The two leads worth carrying forward, stated with their caveats. (1) T1, the declining coefficient path: the differential growth of AI-exposed sectors in high-AIS countries has trended *down* (-0.57 pp/year, p = 0.02 unadjusted, ~0.14 after correcting for the eight-spec trace family) — driven by the 2018 peak-to-2023 trough, i.e. the global tech cycle, not AI; its importance is as a warning that the pre-period trend must be modeled, not assumed flat, when the 2026-27 vintages arrive. **(2) G2, the producer unmasking:** removing AI-producer economies (KOR, TWN, NLD, IRL, SGP) doubles the negative association between AIS and 2025 growth surprises (to -0.57, p = 0.13) — directionally consistent with the popular 'the capex boom pays producers first, adopters later' reading, and exactly the masking pattern the confounder map predicted for producer exposure. Both are hypothesis-generating flags, not findings. Everything else — including the sector splits and the acceleration hypothesis — shows **no trace** in 2025, and the trace map shows every unadjusted band-crossing sits in pre-measurement years.



11b. The white-collar layer (v2.2): a confounder closed, a ratio rule vindicated

The ILOSTAT employment-by-occupation table (user-supplied; ISCO-08, 118 panel countries, mostly 2024-2025 vintages) closes the occupational-mix row of the scoreboard, with an anticlimactic and therefore reassuring result: the white-collar share (ISCO 1-4, per frozen D3) adds essentially nothing to the adoption model once income, size and region are held — coefficient 0.82 (HC1 0.54), R-squared gain 0.007 (WC1) — and the AIS measure is invariant to it (v1-v2 correlation 0.987; the only notable mover is Korea, dropping 16 ranks once its large white-collar workforce is credited). The priority-1 and macro nulls are unchanged under AIS v2 (E1: -2.07, p = 0.46; W1: -0.09, p = 0.85). The feared 'countries with more cognitive workers adopt more' confound is, empirically, almost entirely absorbed by income. Separately, the professional-intensity margin (Axis A) delivered a textbook validation of the design's ratio rule (D2): the raw adoption-per-white-collar-worker ranking is topped by NER, BDI, SEN, TCD, MOZ — low-income economies whose tiny white-collar denominators mechanically inflate the ratio. The residualized version behaves (correlation 0.66 with AIS; W-branch test null, p = 0.70), but the raw ratio goes exactly where the skeleton said ratios go: descriptive use only.

Minimum detectable effects (80 percent power), the design's ceiling. Per one standard deviation of AIS (sd = 0.43 logit): E1 productivity design MDE ~2.4 pp of differential yearly growth against plausible effects of 0.3-1.0 pp — underpowered by roughly 3-8x at a single year; W1 macro design MDE ~0.48 pp of growth surprise against plausible 0.1-0.5 pp — underpowered by roughly 1.5-4x; the diffusion-divergence design is fully powered (and detects decisively). Power grows with cumulated horizons: the 2026-27 vintages roughly halve the E1 MDE by pooling two more outcome years against a frozen treatment.

12. What remains: the TODO roadmap

Item	Blocked on	Expected information value
2026-27 Eurostat + WEO vintages against the frozen AIS (the prospective test)	time	HIGHEST - turns h=0 into h=1-2 and halves the MDE
External registration of the frozen confirmatory specs (OSF-style, hashed)	user action	high - converts self-attestation into verifiable preregistration
Latent-factor / IV provider reconciliation as core AIS (C2)	a second AEI vintage window; OpenAI Signals	high - D2 says single-provider AIS overclaims

ILOSTAT white-collar denominators	DONE (v2.2)	delivered: confounder closed, ratio rule vindicated
Monthly services output (sts_sepr_m) as fast intermediate outcome	user download from Eurostat	medium - adds within-year timing
H6 shock-interaction family	an identified 2025-26 shock series (e.g. tariff/energy shocks)	medium - the second core family, untouched
OpenAI Signals + OpenRouter depth measures	egress or user download	medium - third/fourth provider for reconciliation
Producer-vs-user decomposition done properly (semiconductor trade weights)	trade data (UN Comtrade blocked)	medium - follows the G2 trace
Market-layer downstream validation (country equity returns)	price history source	low by design (most confounded layer)

13. Provenance

Artifact	Bytes	SHA-256 (first 16)
S02 Microsoft AI Diffusion public repository (country CSV)	4468	8e304330f5a63779
S02b Microsoft AI Diffusion repo README (registry fingerprint c	3541	016d900e299594e0
S01 Microsoft Global AI Diffusion Q1 2026 report PDF	1784281	3bf9f87b63372412
S01b Microsoft Global AI Diffusion 2025 H1 report PDF (first re	7746221	a1220d061a69362b
S15 World Bank GDP current US$ (Frictionless datasets/gdp mirr	562767	8d84ef6bcaccf740
X01 World Bank population total (Frictionless datasets/populat	552112	7d2dd6a17f5ed791
X02 Country codes / regions concordance (Frictionless datasets	134003	67b009b529330b0a
S04a AEI v3 raw Claude.ai usage (2025-08-04..11), Sept 2025 rel	18894517	c8ef9c5eee0c42fe
S04b AEI v3 raw 1P API usage (2025-08-04..11), Sept 2025 releas	7027019	c0fc061ea7d4feb7
S04c AEI v3 data documentation (Sept 2025 release)	18245	cc57723797b61a7e
S12 IMF WEO Database October 2024 full country dataset (pre-AI	20297276	46d492250b59b92e
S11 IMF WEO April 2026 vintage, IMF data portal export (WEO 9.	8464348	eb2cfa088e5a168c
S08 Eurostat nama_10_lp_a21 full linear export (labour product	79860740	a38fe552efacd757
S14 ILOSTAT employment by sex and occupation (EMP_TEMP_SEX_OCU	19881535	73685f07c78a5537

Microsoft repository vintage: git HEAD 8fe84a3b797e89846e892592dbb9d0022e9b7dd2. Permutations: 10,000 per spec (2,000 for the family max-stat), seed 20260825. Full run ledger: exploration/search_ledger.json; validation record: manifest/validation.json. Blocked sources are recorded with status blocked_by_egress in manifest/manifest.json. This note was generated by reports/generate_note.py; no numeric result was typed by hand.